Real time intraoperative functional brain mapping using a RGB camera

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ABSTRACT

We demonstrate that a RGB camera and a continuous wave white light illumination is a suitable approach to intraoperatively localize in real time the sensory and motor areas of the patient brain. The analysis of the reflectance spectra of the acquired light and Monte Carlo simulations allow us to measure the concentration changes of oxygenated and deoxygenated hemoglobin during the video acquisition. These measures are compared to the expected hemodynamic response to precisely localize the functional areas of the patient brain.

Keywords: hemodynamic response, functional brain mapping, Monte-Carlo simulations, intraoperative imaging, RGB camera, real time

1. INTRODUCTION

Non-invasive functional brain mapping is an imaging technique used to localize the functional areas of the brain. This technique is used before brain tumor resection surgery to indicate to the neurosurgeon the cortical tissues which should not be removed without cognitive impairment. fMRI is widely used to localize patient sensory or motor areas. However, after patient craniotomy, a brain shift invalidates the relevance of neuro-navigation to intraoperatively localize the functional areas of the patient brain. For this reason, the localization of cortical motor or sensory areas is achieved through electrical stimulation but this technique suffers from limitations.

Chance et al. have demonstrated that optical imaging can be used to monitor brain activity with the determination of concentration changes of oxygenated hemoglobin $\Delta[HbO_2]$, deoxygenated hemoglobin $\Delta[Hb]$, blood volume $\Delta[Hb]+\Delta[HbO_2]$ and oxygenated cytochrome-oxidase.[1] Since then, numerous works have demonstrated the ability of optical imaging to detect functional areas thanks to hemodynamics.[2–10] It also has been demonstrated that a device consisting of a continuous wave white light illumination and a RGB camera is a suitable approach.[11,12]

The objective of this work is to supply the methodological tools for the real time construction of quantitative functional brain maps based on the analysis of the cortical reflectance spectra acquired by a digital RGB camera. The motivation for this paper is derived from the necessity to analyze the hemoglobin time courses in brain tissue following neuronal activation which are closely linked to the BOLD (blood oxygenation level dependent) contrast used in functional MRI studies.[13] Concentration changes of oxygenated hemoglobin and deoxygenated hemoglobin can be measured for each camera pixel using the modified Beer-Lambert law.[14] This law is computed with an estimated mean path length calculated by Monte-Carlo simulations.[15] Finally, the concentration changes time courses are compared to the expected hemodynamic response of the patient brain which allow us to precisely localize the functional tissues of the patient brain.

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2. MATERIAL AND METHODS

2.1 Experimental setup

The instrumentation consisted of a digital RGB CMOS camera (BASLER acA2000- 165uc), a continuous wave white light source (OSRAM Classic 116W 230V Light bulb) and a laptop (processor: Intel Core $i5-7200U$, $2.50GHz \times 4$, ram: $15.3GiB$). The RGB camera was used with an Edmund Optics camera lens ($f = 16mm f\# = 1:1:4$). The sensor provided 8 bits RGB images every 33 ms.

2.2 Patient inclusion and experimental paradigm

The video acquisition was realized before a brain tumor resection surgery. The patient was awakened and functional tests have been realized. Stimulation of the motor cortex was achieved through successive fist clenching and relaxing. The paradigm consisted of 3 steps: 30s of rest, followed by 30s of stimulation and 30s of rest. The neurosurgeon realized electrical brain stimulation to localize patient brain motor and sensory areas.

2.3 Data pre-processing

Once an image is acquired, several steps are processed in parallel. First, the image is registered to compensate the brain repetitive motion. The algorithm developed by *Sdika et al.*[16,17] was implemented to ensure that each camera pixel corresponds to the same cortical area all along data acquisition. The RGB intensities of the acquired image are then independently filtered with a low-pass IIR Bessel temporal filter.[1] This aims to select the low frequencies of the light absorption which are associated with the cortical activity.

2.4 Model

A functional model is applied on the acquired images to process functional brain maps.

2.4.1 Data correction

The slow drift of the collected intensity due to cortical tissue desiccation during the video acquisition[18] was corrected through a linear regression subtracted to the original data.

2.4.2 Functional model

The assessment of the concentration changes through the modified Beer-Lambert law[14] depends on the determination of the mean path length of the detected photons. Monte-carlo simulations[15] were processed to estimate a global mean path length which was applied to all camera pixels. The computed concentration changes times courses were compared to the expected hemodynamic response with the Pearson correlation coefficient. Only pixels whose correlation coefficient was superior to a threshold value was considered. Finally, the quantitative functional map $QMap_C$ was computed by averaging $\Delta C(t)$ over the patient activity period. C represents either Hb or HbO_2. The computation of the quantitative functional maps is summarized in the following equation:

$$QMap_C(x,y) = \begin{cases} \dfrac{\sum_{n=N_1}^{N_2} \Delta[C](n,x,y)}{N_2 - N_1}, & \text{if } r_C(x,y) \geq PCCT \\ \text{non processed}, & \text{otherwise.} \end{cases} \tag{1}$$

(x,y) denotes an image pixel position. n is the frame index, N_1 and N_2 are the patient hand stimulation starting and ending frame indexes. $r_C(x,y)$ denotes the Pearson correlation coefficient calculated between the chromophore C concentration changes time course at the image pixel position (x,y) and the expected hemodynamic response. A 5×5 median filter is applied to the quantitative functional maps for cosmetic purposes.

2.5 Real time processing

These processing steps are implemented in a real-time architecture. In our application, the term real time means that the functional brain maps are computed during the data acquisition, a few seconds after the start of the patient physiological activity. Indeed, following the neuronal activation, a slow oxygenation change occurs in the activated area. The functional maps cannot be computed for each new acquired image but few seconds after the start of the patient physiological stimulus. The pre-processing steps (see Sec. 2.3) and the model (see Sec. 2.4) are executed in different parallel threads. An intermediate buffer is placed between each computational step allowing the data storage. These buffers are required in our real-time architecture since the processing times are different for each computational step. The thread corresponding to the model is computed as soon as the cortical activity started. This thread is recursively called while images are pre-processed. The real time computation of the functional brain maps is represented in the Fig. 1.

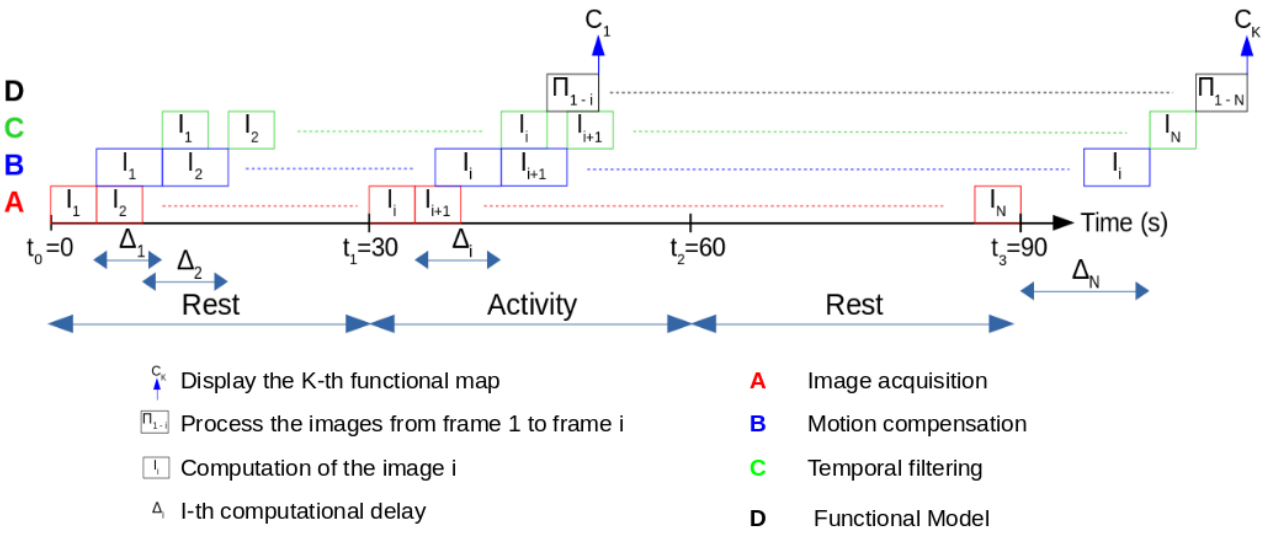

Figure 1. Overview of the real time processing. The letters from A to D designate the different thread executed in parallel. The rectangle length represents a thread computation time. For each new acquired image, the brain motion is compensated in the thread B. Then, the RGB intensities of the acquired image are independently filtered in the thread C. Once the cortical activation started, the functional maps are recursively computed in the thread D until the end of the acquisition.

3. RESULTS AND DISCUSSION

In Fig. 2, the real time *Hb* quantitative functional maps (see Sec. 2.4) which are best correlated with the expected hemodynamic response are plotted (correlation threshold sets to 0.5). The highest *Hb* mean concentrations are localized on blood vessels surrounding motor and sensory areas. The highest correlation coefficient are localized on these blood vessels and on motor and sensory areas. The computed functional maps become significant as soon as the patient activity is established, see Fig. 2−3. *Pichette et al.*[19] proposed to mask the blood vessels in order to render more clearly the smaller hemodynamic changes in grey matter. The measurement of the correlation coefficient between the concentration change time courses and the expected hemodynamic response allows us to visualize the activated cortical areas without masking the blood vessels. A global mean path length is used for each camera pixels in the modified Beer Lambert law.[14] A pixel-wise determination of the mean path length could be implemented to assess more precisely the *Hb* concentration changes. However it would require a lot of Monte Carlo simulations and cause an increase of the processing time of the model (see Sec. 2.4). The slow drift of the collected intensity is corrected each time the model is executed (see Sec. 2.4) which cause an increase of the processing time. In addition, the estimation of the collected intensity slope is not optimal for the first maps (from Fig. 2−1 to Fig. 2−7). This problem will be investigated in a future work.

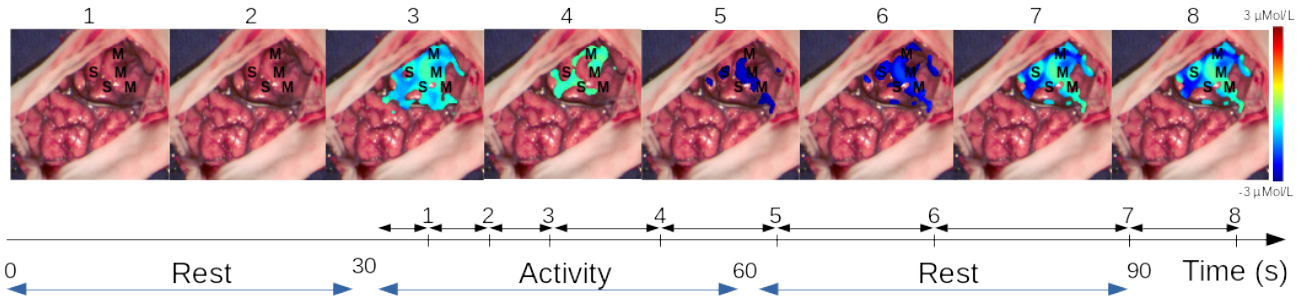

Figure 2. *Hb* concentration changes maps emitted during the video acquisition. The correlation threshold is set to 0.5. The horizontal axis represents the acquisition time of the RGB camera. The experimental paradigm is represented under this axis. The vertical lines represent the display time of the computed functional maps (images from A to H). The double black arrows represent the processing time of the model, see Sec. 2.4. The markers **M** and **S** designate respectively the motor and the sensory areas which have been validated by the neurosurgeon through electrical brain stimulation. The colorbar represents the scale of variation of *Hb* mean concentration in $\mu Mol/L$.

4. CONCLUSION

In conclusion, a RGB camera and a continuous wave white light illumination seem to be a suitable setup to intraoperatively monitor the sensory and motor areas of the patient brain in real time. These results indicate that our setup could complement the actual intraoperative gold standard for the localization of patient functional areas.

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